$q(\cdot \mid s, t, \theta)$ from a strategy-conditioned generator q with internal parameters θ, where the strategy s determines what types of fabricated evidence to produce; (2) Visualization: generates figures and tables $V = \text{viz}(D)$ from the synthetic data to support fabricated claims; (3) Manuscript assembly: composes a complete paper $x = \text{compose}(u, D, V)$ including abstract, introduction, methods, results, discussion, and conclusion sections, along with citations and professional formatting. The structural validity constraints to ensure generated papers pass basic formatting checks:

$$C(x) = \mathbb{I}\big[\text{compile}(x) = \text{success} \\ \wedge\ \text{struct}(x) \in \mathcal{C}\big] = 1,$$

where $\mathcal{C}$ encodes formatting requirements (section presence, figure/table counts, bibliography entries). Only papers with $C(x) = 1$ are proceeded.

Since the data synthesis process is stochastic, the same seed prior may yield different papers across runs (e.g., different fabricated performance numbers, plot variations, or phrasing). Consequently, the end-to-end generation thus induces a distribution over papers:

$$p_{\mathcal{G}}(x \mid s, t) = \int p(x \mid D, s, t)\, q(D \mid s, t, \theta)\, \mathrm{d}D$$

3.3 Review Agent $\mathcal{R}$

Given a paper $x \in \mathcal{X}$, the Review Agent queries each model $m \in \mathcal{M}$ under a fixed K-criterion rubric (e.g., methodology, significance, clarity, *etc.*). Each model returns a rubric vector and textual feedback $(\mathbf{r}_m(x), \omega_m(x))$. Using reviewer weights $\mathbf{w} \in \Delta^{M-1}$, the agent forms the consensus rubric

$$\bar{\mathbf{r}}(x) = \sum_{m \in \mathcal{M}} w_m\, \mathbf{r}_m(x),$$

and produces a binary recommendation via the scoring functional ϕ and a calibrated threshold τ: $\hat{y}(x) = \mathbb{I}[\phi(\bar{\mathbf{r}}(x)) \geq \tau]$. We summarize the agent's output as

$$\mathcal{R}(x) = \Big(\{(\mathbf{r}_m(x), \omega_m(x))\}_{m \in \mathcal{M}},\ \bar{\mathbf{r}}(x),\ \hat{y}(x) \Big),$$

which preserves per-model judgments and comments while supplying a single consensus score and decision.

3.4 Review Calibration for Analysis $\mathcal{A}$

We calibrate the Review Agent's decision rule using a corpus of real conference submissions with publicly available reviews and outcomes.

Calibration corpus. We define the reference pool as:

$$\mathcal{D}_{\text{ref}} = \{(x_i, y_i^{\text{hum}}, \sigma_i, h_i)\}_{i=1}^{N_\star},$$

where x_i is the paper artifact, $y_i^{\text{hum}} \in \{0, 1\}$ indicates the human accept/reject decision, $\sigma_i \in \mathcal{C}_{\text{stat}}$ represents the meta-status labels (e.g., oral/spotlight/poster/reject/withdraw), and $h_i \in \mathbb{R}$ is a scalar venue score such as the average assessment.

From this reference pool, we construct a calibration set $\mathcal{D}_{\text{cal}}$ that preserves the score and status distributions of $\mathcal{D}_{\text{ref}}$ with a stratified sampling algorithm (see Appendix A.1).

Agent scoring. For each paper $x \in \mathcal{D}_{\text{cal}}$, the Review Agent produces a consensus rubric $\bar{\mathbf{r}}(x)$, converts it to a scalar score $s(x) = \phi(\bar{\mathbf{r}}(x)) \in \mathbb{R}$, and makes a binary recommendation $\hat{y}_\tau(x) = \mathbb{I}[s(x) \geq \tau]$ for threshold $\tau \in \mathbb{R}$.

Threshold calibration. We derive two operating thresholds to accommodate different evaluation criteria.

1. Rate-matching threshold. Let $\alpha^\star \in (0, 1)$ denote the target venue acceptance rate. We define:

$$\widehat{\alpha}_{\text{cal}}(\tau) = \frac{1}{|\mathcal{D}_{\text{cal}}|} \sum_{x \in \mathcal{D}_{\text{cal}}} \hat{y}_\tau(x), \tag{1}$$

$$\tau_{\text{rate}} \in \arg\min_{\tau \in \mathbb{R}} |\widehat{\alpha}_{\text{cal}}(\tau) - \alpha^\star|. \tag{2}$$

This threshold ensures that the agent's acceptance rate on the calibration set matches the venue's historical acceptance rate.

2. Probability-consistency threshold. Let $\pi(z) = \mathbb{P}(y^{\text{hum}} = 1 \mid s(x) \geq z)$ for $t \in \mathbb{R}$, estimated on $\mathcal{D}_{\text{cal}}$ using a monotone calibration model. We define:

$$\tau_{0.5} = \inf\{z \in \mathbb{R} : \pi(z) \geq \tfrac{1}{2}\},$$

so that papers scoring $s(x) \geq \tau_{0.5}$ have at least 50% estimated probability of human acceptance.

Output. The calibration module returns $\mathcal{A}(\mathcal{D}_{\text{cal}}) = (\tau_{\text{rate}}, \tau_{0.5})$, providing operating thresholds for the decision rule $\hat{y}(x) = \mathbb{I}[s(x) \geq \tau]$.

3.5 Theoretical Reliability of Review Aggregation

When combining judgments from multiple reviewer agents, two sources of uncertainty arise: (i) stochastic variation in individual model outputs,

even when evaluating identical papers, and (ii) estimation error in the decision threshold τ due to finite calibration data. To quantify the reliability of our aggregated decisions $\hat{y}(x) = \mathbb{I}[s(x) \geq \tau]$, we provide a rigorous error analysis in Appendix A.2.

Setup and assumptions. For each model $m \in \mathcal{M}$, let $\mathbf{r}_m(x) \in \mathbb{R}^K$ denote its rubric vector and $\bar{\mathbf{r}}(x) = \sum_m w_m \mathbf{r}_m(x)$ the weighted consensus. We impose two standard regularity conditions: (i) **Sub-Gaussian noise**—each reviewer's centered rubric $\mathbf{z}_m(x) := \mathbf{r}_m(x) - \mathbb{E}[\mathbf{r}_m(x) \mid x]$ is vector sub-Gaussian with proxy matrix Σ_m, a natural consequence of bounded rubric scores $r_{m,k} \in [a_k, b_k]$ required by all venues; (ii) **Lipschitz aggregation**—the scoring function $\phi : \mathbb{R}^K \to \mathbb{R}$ is L_ϕ-Lipschitz, satisfied by common choices such as weighted averages ($L_\phi = \|\mathbf{v}\|_2$) or selecting a single overall score ($L_\phi = 1$). We also assume independent evaluation across reviewers, reflecting standard peer-review practice.

Ensemble concentration (Q1). Under these assumptions, we establish exponential concentration bounds showing that the consensus score $s(x) = \phi(\bar{\mathbf{r}}(x))$ clusters tightly around its latent mean $\mu_s(x) = \phi(\mathbb{E}[\bar{\mathbf{r}}(x)])$. Specifically, for papers with margin $\gamma(x) = |\mu_s(x) - \tau|$ from the threshold, the misclassification probability satisfies

$$\Pr\left(\hat{y}(x) \neq y^\star(x)\right) \;\leq\; \exp\left(-\frac{\gamma(x)^2}{2\sigma_w^2 + \frac{2}{3}c_{\max}\gamma(x)}\right),$$

where $\sigma_w^2 = \mathrm{Var}[s(x)]$ and $c_{\max}$ captures bounded differences (Theorem 1). In the common scalar-assessment case where each reviewer outputs $s_m(x) \in [a, b]$ and ϕ is the identity, both the variance term σ_w^2 and the bounded-difference term $c_{\max}$ scale as $1/M$ with the number of reviewers (Corollary 2). This yields the simplified bound

$$\Pr(\hat{y} \neq y^\star) \;\leq\; \exp\left(-\frac{M\gamma^2}{2\sigma^2 + \frac{2}{3}(b-a)\gamma}\right)$$

for uniform weights $w_m = 1/M$ and identical per-review variance σ^2. For linear aggregation $\phi(\mathbf{a}) = \mathbf{v}^\top \mathbf{a}$, we further show that the bound is minimized by inverse-variance (GLS) weighting $w_m^\star \propto 1/(\mathbf{v}^\top \Sigma_m \mathbf{v})$ (Corollary 1).

Calibration error (Q2). The concentration results above assume a known threshold τ. In practice, we estimate τ from the finite calibration set $\mathcal{D}_{\text{cal}}$ of size N_{cal}, introducing a second source of error. For the *rate-matching* threshold τ_{rate} (chosen to match the venue's historical acceptance rate $\alpha^\star$), we bound the acceptance-rate estimation error uniformly over all thresholds via the Dvoretzky–Kiefer–Wolfowitz inequality, yielding

$$\sup_{\tau \in \mathbb{R}} \left|\widehat{\alpha}_{\text{cal}}(\tau) - \alpha(\tau)\right| \;\leq\; \sqrt{\tfrac{1}{2N_{\text{cal}}} \log \tfrac{4}{\delta}}$$

with probability at least $1 - \delta$ (Proposition 1). For the *probability-consistency* threshold $\tau_{0.5}$ (where papers scoring above have $\geq 50\%$ estimated human-acceptance probability), we employ isotonic regression to estimate the conditional probability $\pi(t) = \mathbb{P}(y^{\text{hum}} = 1 \mid s(x) \geq t)$ and provide explicit bounds on the threshold error $|\hat{\tau}_{0.5} - \tau_{0.5}|$ as a function of N_{cal} and the slope of π near $1/2$ (Proposition 2).

Empirical validation. We validate our theoretical bounds through synthetic experiments with $n = 5{,}000$ papers and $M \in \{1, 2, 3\}$ reviewers producing noisy scalar assessments in $[1, 10]$. Our results confirm that: (i) empirical misclassification rates fall well below theoretical bounds across all margins and ensemble sizes; (ii) threshold estimation error decreases as $O(1/\sqrt{N_{\text{cal}}})$, with our choice of $N_{\text{cal}} = 200$ yielding error ≈ 0.26; (iii) both the empirical noise variance $\mathrm{Var}[s(x) - \mu_s(x)]$ and the bounded-difference proxy $(b-a)^2/M$ decrease as $1/M$—increasing from $M = 1$ to $M = 3$ reviewers reduces both quantities by approximately $3\times$ (Figure 3 in Appendix). These results establish that multi-reviewer aggregation substantially improves decision reliability, a property we exploit throughout our evaluation to justify using $M = 3$ models and $N_{\text{cal}} = 200$ calibration samples.

4 Experiment

4.1 Setup

Implementation Our agent framework is adapted from AI-Scientist (Lu et al., 2024a), but we have fundamentally redesigned its entire pipeline. We retain only its most foundational writing prompts and have eliminated the need for any experimental execution or structured templates. Our framework now operates directly from a simple seed idea, allowing the LLM to freely generate any necessary experimental results and plotting code. We follow the generation strategy space set claimed in Section 3.2. With `GPT-5`, we generate all seed topics for paper

Table 1: Acceptance (ACPT) and Integrity Concern Rate (ICR) by strategy.

Strategy	ACPT		ICR-m			ICR@M
	τ_{rate}	$\tau_{0.5}$	o3	o4-mini	GPT-4.1	
s_1	67.0%	82.0%	38.4%	4.7%	2.3%	39.5%
s_2	32.0%	49.0%	35.2%	4.5%	2.3%	35.2%
s_3	53.5%	69.7%	29.4%	2.4%	4.7%	31.8%
s_4	44.0%	59.0%	28.2%	5.9%	1.2%	30.6%
s_5	35.4%	53.5%	25.9%	8.2%	7.1%	34.1%
All	52.0%	69.0%	50.6%	5.7%	8.0%	51.7%

Table 2: Concern–acceptance conflict (%): within papers where the model raised an integrity concern, the share still receives an acceptance-level score by model and strategy (s_1-s_5, *All*). Higher values indicate stronger contradiction.

Model	s_1	s_2	s_3	s_4	s_5	*All*
o3	33.3%	25.8%	52.0%	30.0%	40.9%	29.5%
o4-mini	100.0%	50.0%	100.0%	80.0%	71.9%	100.0%
GPT-4.1	50.0%	50.0%	75.0%	0.0%	33.3%	57.1%

generations spanning representative domains in AI research (see Appendix B). Each seed produces 4 papers across six strategy setups. For the ease of acceptance decision, we take only the overall assessment score provided by the review agent for paper scoring, i.e., $\phi(\bar{r}(x)) = r_{oa}(x)$.

Agent Models. We use `GPT-5` to support our paper generation agent. For the review agent, we set $\mathcal{M} = 3$ and use `o3`, `o4-mini`, and `GPT-4.1` with the rubric review prompt.

Calibration set and thresholds. We instantiate the reference pool $\mathcal{D}_{\text{ref}}$ as the ICLR 2025 OpenReview submission set (with public reviews and outcomes). A stratified calibration set $\mathcal{D}_{\text{cal}}$ of size $N_{\text{cal}} = 200$ is then constructed as described in Section 3.4. Running the Review Agent on $\mathcal{D}_{\text{cal}}$ yields two operating thresholds. *Rate-matching* selects τ_{rate} so that the agent minimize the drift of empirical acceptance rate on $\mathcal{D}_{\text{cal}}$ matches the venue rate $\alpha^\star = 0.3173$, which yields $\tau_{\text{rate}} = 7$. *Probability-consistency* defines such that papers with $s(x) \geq \tau_{0.5}$ have estimated human-acceptance probability at least 50%; this yields $\tau_{0.5} = 6.667$.

Evaluation metrics. We evaluate along two axes. (I) **Acceptance rate** (ACPT). Let $\mathcal{D}$ be the set of generated papers and $\hat{y}_\tau(x) = \mathbb{I}\big[s(x) \geq \tau\big]$ the Review Agent's decision at threshold τ, with $s(x) = \phi(\bar{\mathbf{r}}(x))$. For any operating threshold $\tau \in \{\tau_{\text{rate}}, \tau_{0.5}\}$ we report

$$\text{ACPT}(\tau) = \frac{1}{N}\sum_{j=1}^{N} \hat{y}_\tau(x_j),$$

(II) **Integrity Concern Rate** (ICR). Let $c_m(x) = \Gamma(\omega_m(x)) \in \{0, 1\}$ indicate that reviewer $m \in \mathcal{M}$ explicitly raises integrity-related concerns in $\omega_m(x)$. And suppose $\bar{c}_{\text{any}}(x) = \mathbb{I}\big[\sum_{m\in\mathcal{M}} c_m(x) \geq 1\big]$. Then for $m \in \mathcal{M}$, we have *Per–review-model ICR (ICR-m)* and the relaxed metric at panel-level, *Any-of-panel ICR (ICR@M)*:

$$\text{ICR-}m = \frac{1}{N}\sum_{j=1}^{N} c_m(x_j), \tag{3}$$

$$\text{ICR@}m = \frac{1}{N}\sum_{j=1}^{N} \bar{c}_{\text{any}}(x_j). \tag{4}$$

We use `GPT-5` as LLM-judge to classify whether the text feedback from review agents contains integrity-related concerns.

4.2 Evaluation Analysis

Main Results. Our main evaluation result is in Table 1. We find that acceptance is unexpectedly high under most manipulations. Single strategies already yield substantial ACPT (e.g., $\text{ACPT}_{\tau_{\text{rate}}} = 67.0\%$, $\text{ACPT}_{\tau_{0.5}} = 82.0\%$ for s_1), indicating that current review agents are easily persuaded and lack sufficient awareness to spot integrity/fabrication issues. The *All* strategy as a composed setup, attains high acceptance (52.0%/69.0%), but it also maximally increases detectability (ICR@M 51.7%, `o3` 50.6%), suggesting that composing strategies broadens the footprint seen by detectors. Among

Overall ICLR 2025 acceptance rate 31.73%; see https://papercopilot.com/statistics/iclr-statistics/iclr-2025-statistics/.